👩‍🏫 Staff Guide · Biology A · Biofuel & Energy (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Ms. Dunham's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
A3 Matter & Energy	Reading parts 1-2; notes section A
A5 Making Lipids	Reading part 3; notes section B
A6 Biofuel + Combustion lab	Reading part 4; notes section B
A7 Energy Reactions; A8 Carbon Cycle; A10	Reading part 5; notes section C
A2.10 Review + Algae Biofuel Final Exam	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Ms. Dunham's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Ms. Dunham.

⛽ Biofuel & Energy: Guided Notes

Biology A · Unit 3 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A3 Matter & Energy, A5 Lipids, A6 Biofuel + Combustion, A7 Energy Reactions, A8 Carbon Cycle, Final Exam

Name: _____

Use the class slides and your reading. Key words:

matter · *energy* · *potential* · *kinetic* · *lipid* · *biodiesel* · *combustion* · *exothermic* · *endothermic* · *renewable* · *carbon cycle*

A. Matter and energy (A3)

1. Matter has _____ and takes up space.

2. Fill in the table.

Example	Potential or kinetic?
water behind a dam	
a boulder rolling downhill	
gas in a car's tank	

3. A hot pan warms your hand. Is this a transfer or a transformation? Explain.

B. Lipids and biofuel (A5, A6)

4. A fat is glycerol plus three fatty _____.

5. Lipids store about _____ times the energy of sugar.

6. List the energy changes when a bus burns biodiesel.

C. Reactions and the carbon cycle (A7, A8)

7. A reaction that gets hotter is _____.

8. Coal, oil and natural gas are _____ fuels.

9. Name one process that takes CO2 out of the air and one that puts it back.

⛽ Biofuel & Energy: Reading

Biology A · Unit 3 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A3 Matter & Energy, A5 Lipids, A6 Biofuel + Combustion, A7 Energy Reactions, A8 Carbon Cycle, Final Exam

Name: _____

From Algae Oil to Fuel

Matter is anything that has mass and takes up space, like oil, water and air. Energy has no mass; it is the ability to make things move or change. Potential energy is stored energy, like water behind a dam. Kinetic energy is the energy of motion, like water rushing out of the dam.

Energy can be transferred, which means it moves to a new place (a hot pan warms your hand). It can also be transformed, which means it changes form (fuel burns and becomes heat). Energy is never made or destroyed, but some always turns into wasted heat.

Fats and oils are lipids. A fat is made of glycerol plus three fatty acids. Lipids store about 2.5 times more energy than sugar, gram for gram. Algae change sugar into oil to store energy, and that oil can be made into biodiesel.

To make biodiesel, oil is mixed with methanol and a catalyst, which makes biodiesel and glycerol. Methanol and the catalyst are dangerous, so only the teacher handles them. When fuel burns (combustion), fuel + oxygen → carbon dioxide + water + energy. In a bus, chemical energy becomes heat and then mechanical energy that moves the bus.

Some reactions are exothermic: they release heat and the temperature goes up. Others are endothermic: they take in heat and the temperature goes down, like baking soda and vinegar. Biodiesel is renewable because algae and plants grow back fast. Fossil fuels (coal, oil, natural gas) take millions of years to form. In the carbon cycle, plants take CO2 out of the air, and burning and breathing put it back.

Check your reading

1. What is the difference between energy transfer and energy transformation?

2. Why is biodiesel called renewable?

3. Baking soda and vinegar got colder. Which kind of reaction is this?

⛽ Biofuel & Energy: Quick Check

Biology A · Unit 3 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A3 Matter & Energy, A5 Lipids, A6 Biofuel + Combustion, A7 Energy Reactions, A8 Carbon Cycle, Final Exam

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which is matter?

A. Oil | B. Heat | C. Light

2. Water rushing out of a dam has…

A. No energy | B. Only potential energy | C. Kinetic energy

3. Lipids store about how much energy compared to sugar?

A. Half as much | B. The same | C. 2.5 times more

4. A reaction that takes in heat is…

A. Exothermic | B. Endothermic | C. Renewable

5. Write the word equation for combustion.

6. Is algae biodiesel better for the carbon cycle than oil? Give one reason.

⛽ Biofuel & Energy: Guided Notes · ANSWER KEY

Biology A · Unit 3 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A3 Matter & Energy, A5 Lipids, A6 Biofuel + Combustion, A7 Energy Reactions, A8 Carbon Cycle, Final Exam

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A. Matter and energy (A3)

1. Matter has **mass** and takes up space.

2. Fill in the table.

Example	Potential or kinetic?
water behind a dam	**potential**
a boulder rolling downhill	**kinetic**
gas in a car's tank	**potential (chemical)**

3. A hot pan warms your hand. Is this a transfer or a transformation? Explain.
Transfer: heat energy moves from the pan to your hand without changing form.

B. Lipids and biofuel (A5, A6)

4. A fat is glycerol plus three fatty **acids**.

5. Lipids store about **2.5** times the energy of sugar.

6. List the energy changes when a bus burns biodiesel.
Chemical energy in the fuel → heat (thermal) → mechanical energy that moves the bus.

C. Reactions and the carbon cycle (A7, A8)

7. A reaction that gets hotter is **exothermic**.

8. Coal, oil and natural gas are **fossil** fuels.

9. Name one process that takes CO2 out of the air and one that puts it back.
Out: photosynthesis. Back: burning (combustion) or breathing (respiration).

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

⛽ Biofuel & Energy: Reading · ANSWER KEY

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Check your reading

1. What is the difference between energy transfer and energy transformation?
Transfer: energy moves to a new place. Transformation: energy changes form.

2. Why is biodiesel called renewable?
It is made from plants/algae, which grow back in a short time.

3. Baking soda and vinegar got colder. Which kind of reaction is this?
Endothermic.

⛽ Biofuel & Energy: Quick Check · ANSWER KEY

Biology A · Unit 3 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A3 Matter & Energy, A5 Lipids, A6 Biofuel + Combustion, A7 Energy Reactions, A8 Carbon Cycle, Final Exam

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4. A reaction that takes in heat is…

A. Exothermic | **B. Endothermic** | C. Renewable

5. Write the word equation for combustion.

Fuel + oxygen → carbon dioxide + water + energy.

6. Is algae biodiesel better for the carbon cycle than oil? Give one reason.

Algae take CO2 out of the air as they grow, and they grow back fast (renewable); oil adds old carbon.

🔑 *Key written by our team from Ms. Dunham's quizzes and notes, please check.*